Lab 7

In[1]:=

```
(*Pendulum Equation from Lab 5*)

(*Initialize Omega, l, g*)
ClearAll["Global`*"]
ClearAll[t]
l = 10
g = 980
ω0 = Sqrt[(g/l)]


(*Call NDSolve[]*)
sol = NDSolve[
  {a''[t] == -ω0 ^ 2 Sin[a[t]],
   a'[0] == 0,
   a[0] == 90 Degree},
  a, {t, 0, 2}]

Plot[Evaluate[a[t] /. sol], {t, 0, 2},
  AxesLabel -> {"t (s)", "φ (rad)"},
  PlotLabel -> "Pendulum, φ0 = 90°"]

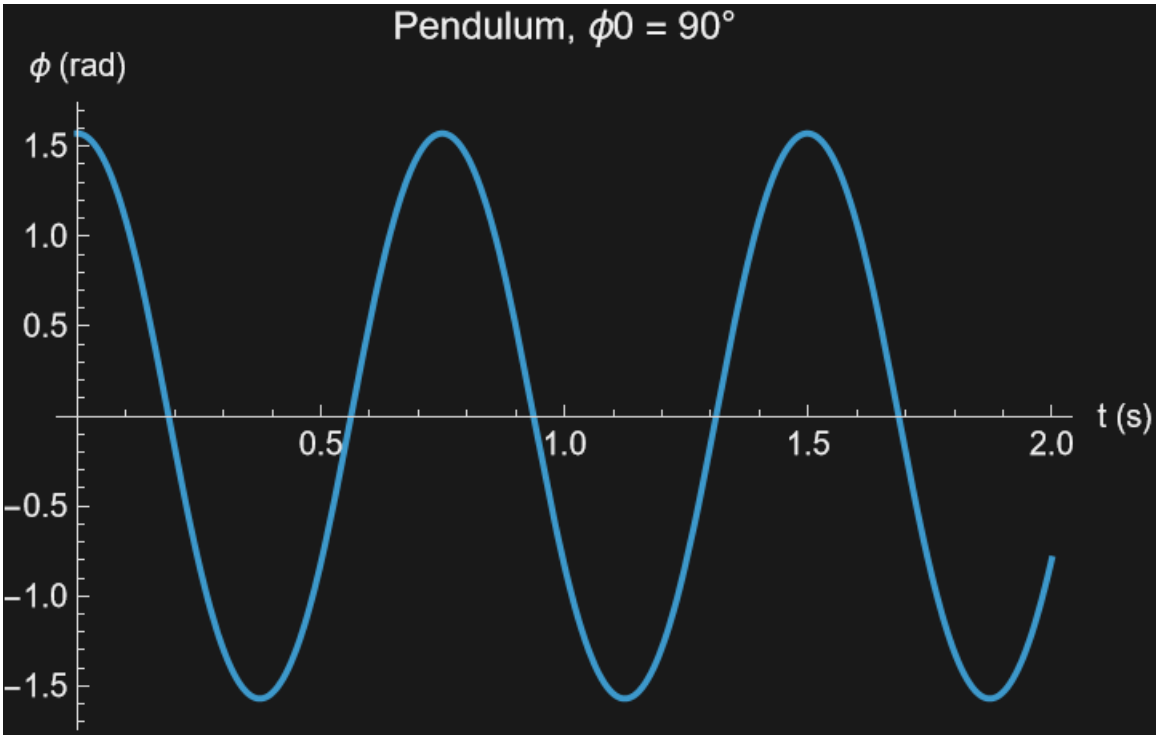
(*Evaluate Function at a Specified Time*)
sol[[1]][[1]][[2]][0.25]
```

10

980

$7\sqrt{2}$

$\{\{a \to \text{InterpolatingFunction}[\langle\langle \text{Domain: } \{\{0., 2.\}\} \text{ Output: scalar} \rangle\rangle]\}\}$



-0.824563

In[2]:=

```
(*Question 1: Figure of Merit*)

(*Define the Analytical Solution*)
ClearAll["Global`*"]
ClearAll[analytical]

φ0 = 30 Degree

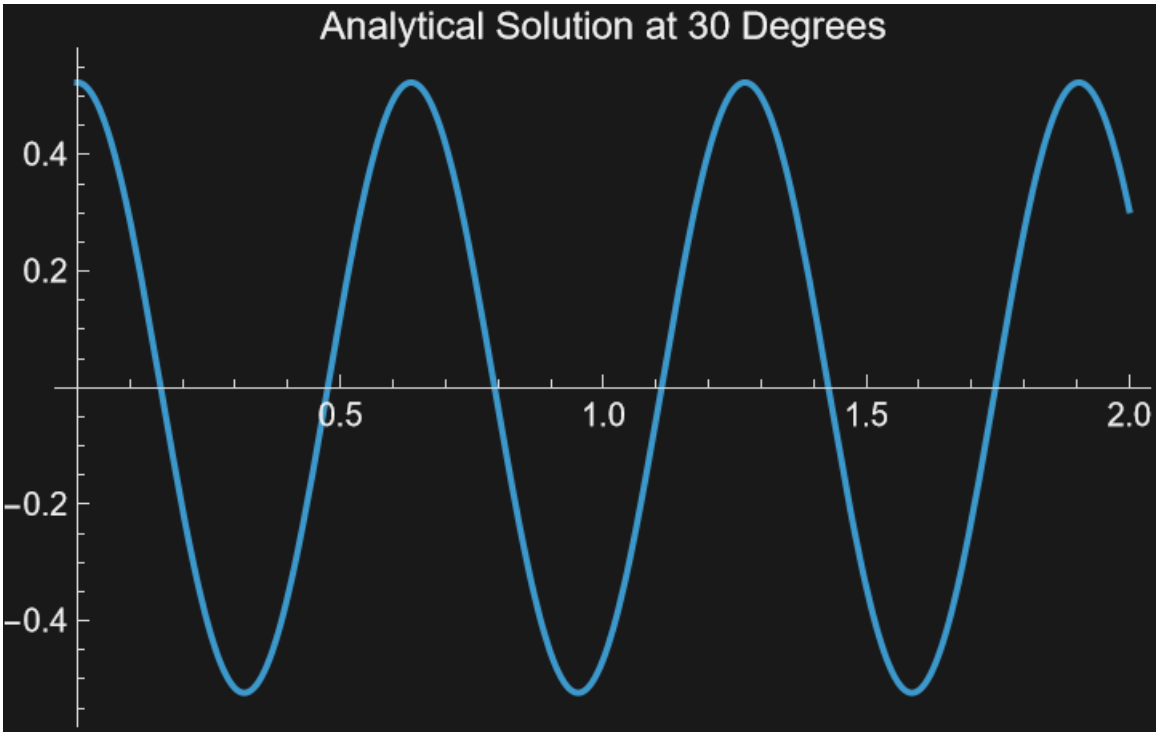
analytical[t_] := φ0*Cos[ω0*t]
Plot[analytical[t], {t, 0, 2}, PlotLabel -> "Analytical Solution at 30 Degrees"]

analytical[0.25]

(*Getting The FOM*)
FOM = Sqrt[NIntegrate[(sol[[1]][[1]][[2]][t] - analytical[t])^2, {t, 0, 2}]]

(*For given angle 30 degrees, the simulation and analytical solution produce very similar
plots.*)
(*For our Figure of Merit (FOM), the closer the value is to zero, the more accruate our
simulated solution is to the analytical one.*)
```

30 °




```
-0.411473
```

```
1.69188
```

In[3]:=

```
(*Question 2: Increasing the Initial Angle*)
ClearAll["Global`*"]
ClearAll[analytical]
ClearAll[φ0]

φ0 = 100 Degree

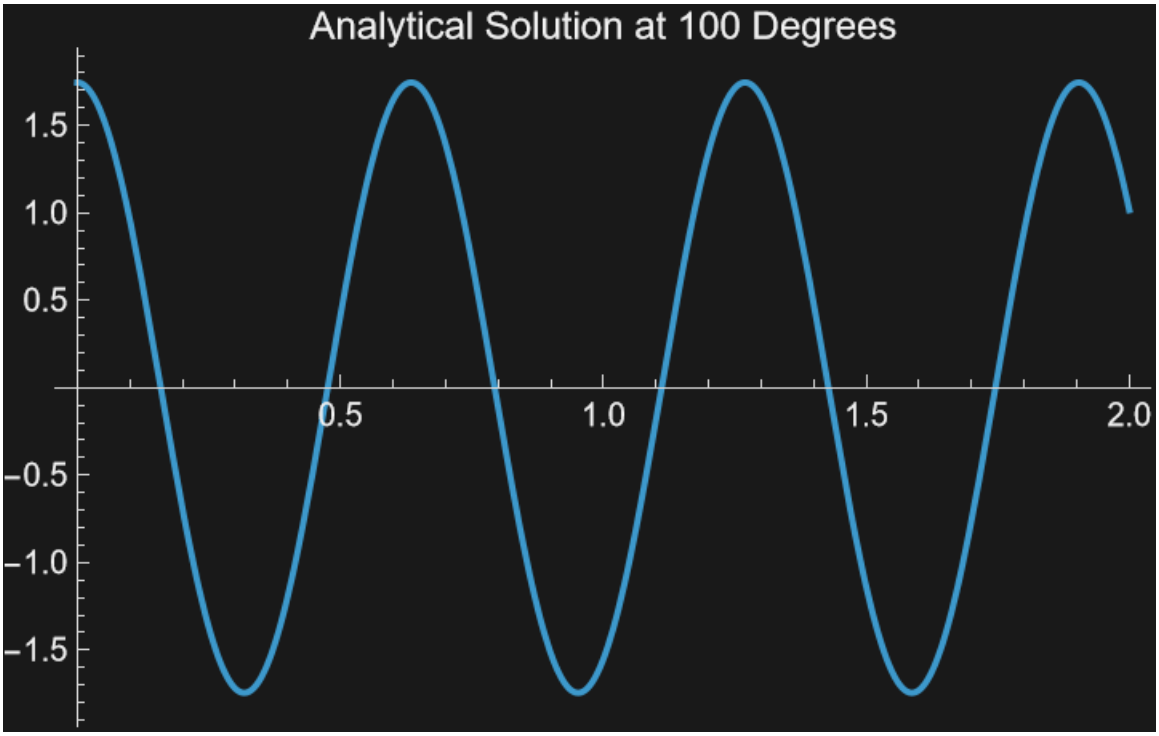
analytical[t_] := φ0*Cos[ω0*t]
Plot[analytical[t], {t, 0, 2}, PlotLabel -> "Analytical Solution at 100 Degrees"]

analytical[0.25]

(*Getting The FOM*)
FOM = Sqrt[NIntegrate[(sol[[1]][[1]][[2]][t] - analytical[t])^2, {t, 0, 2}]]

(*As I increase the initial angle, my simulation results relative to my analytical results
experience decreased accuracy (when observing the FOM).*)
```

```
100 °
```



-1.37158

2.3762

In[4]:=

```
(*Question 3: Plotting the FOM*)
(*Evaluate the FOM at 5 different initial angles and plot them*)

ClearAll["Global`*"]
ClearAll[analytical]
ClearAll[φ0]
ClearAll[FOM]

initialAngles = {}
FOMs = {}

analytical[t_] := φ0*Cos[ω0*t]

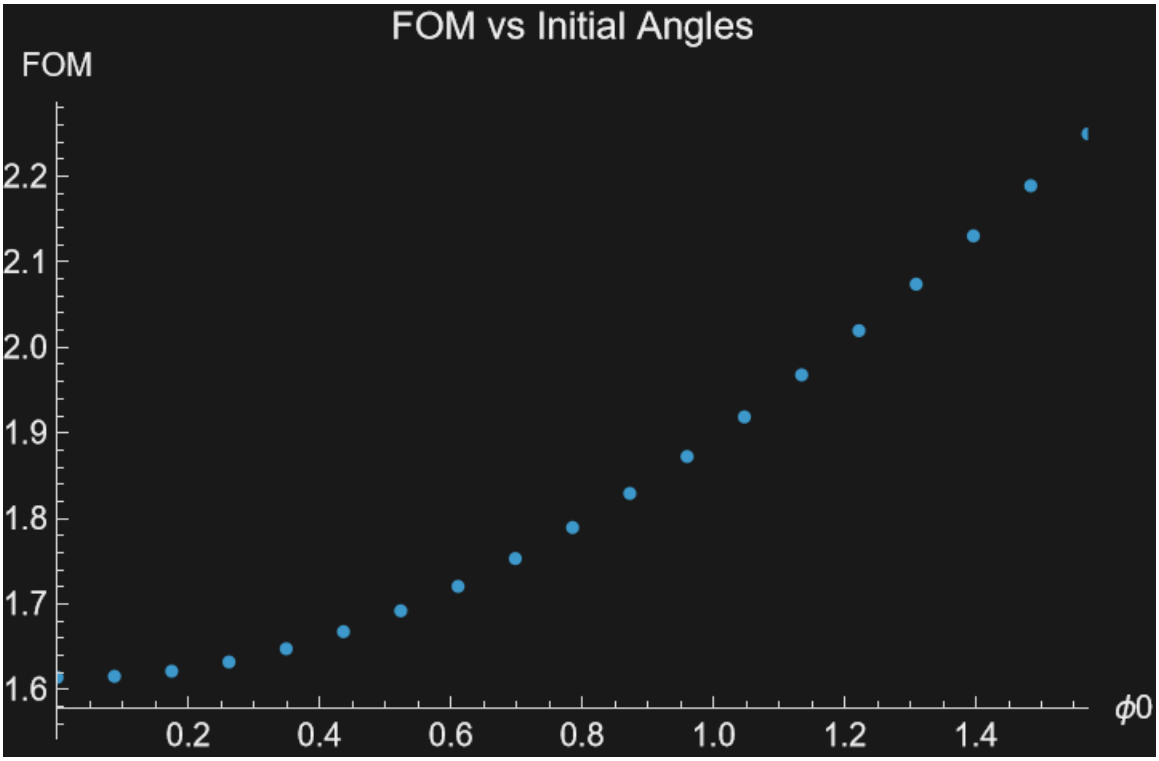
For[i = 0, i <= 90, i += 5,
    φ0 = i Degree;
    AppendTo[initialAngles, φ0];
    FOM = Sqrt[NIntegrate[(sol[[1]][[1]][[2]][t] - analytical[t])^2, {t, 0, 2}]];
    AppendTo[FOMs, FOM];
]

ListPlot[Transpose[{initialAngles, FOMs}],
    PlotRange -> {{0 Degree, 90 Degree}, Automatic},
    PlotLabel -> "FOM vs Initial Angles",
    AxesLabel -> {"φ0", "FOM"}
]

(*I have come to the conclusion that the small angle approximation starts to significantly
deviate (by about 10%) after 40 degrees*)
```

{}

{}



In[5]:=

```
(*Question 4: Increase the Mass of the Pendulum Bob*)
(*Increasing the mass of the pendulum bob wouldn't change anything in our results.*)
(*Recall that we never even initialized a mass 'm' nor used a mass in any of our function
definitions*)
(*The system that we're dealing with here doesn't have mass as an important variable*)
```

4

In[6]:=

```
test = 2 + 2
```

4